***Motivation to Pursue Science and Future Goals*:** I discovered my enthusiasm for basic research at **Grinnell College.** I was recruited for the **Grinnell Science Project**, a summer program designed to inspire students from underrepresented groups to pursue STEM careers. During a one-on-one biology workshop, I learned to fluorescently label actin in *C. elegans,* which peaked interest in the cytoskeleton. By staining F-actin with fluorescein-conjugated phalloidin, I discovered how the dynamic architecture of the cell can drive cell movement and dictate cell polarity. Through additional supportive, inquiry-based learning workshops in biology, chemistry and physics, I grew to appreciate the molecular machinery that underlie life processes. I became motivated to pursue a major in biochemistry, in the hopes of working at the interface of biology and chemistry to understand the macromolecular molecules that drive these unique, unparalleled molecular networks.

My long-term goal is to become a professor of biochemistry, where I will **teach** courses on physical biochemistry and bioorganic chemistry and **conduct research on the structural mechanics of membrane biology** in an **interdisciplinary** academic environment. My hope is to characterize the molecular structures that trigger signaling cascades that lead to diseases that disproportionately affect disadvantaged groups, including HIV/AIDS, diabetes, and cardiovascular disease. It is my vision to become a leader in the fields of membrane and molecular structural biology by using the most innovative and cutting-edge techniques to understand the dynamic architecture of the plasma membrane. As a future investigator, I plan to **mentor** others to not only share my research goals but also my vision to create a more **diverse, empowered** scientific workforce.

Intellectual Merit: From my very first class at Grinnell, I was encouraged to read, write and critically evaluate scientific literature while introducing me to a diverse range of model systems. My small-scale class research projects ranged from studying the synergistic effect of motor oil and water mold (*Saprolegnia sp.*) on *Xenopus laevis* embryos to examining neocentromere (neoCEN) formation in *C. albicans.* These engaging exercises inspired me to pursue full-time research opportunities. I was selected for the **2012 Amgen Scholars Program** at the **University of California, San Francisco (UCSF)** to pursue cutting-edge research with **Dr. Raul Andino.** Here, I was introduced to the dynamic host-virus interface as a model to study cell signaling in the hopes of developing vaccinations and/or antiviral drugs to destabilize aggressive dsRNA viruses, specifically the rotavirus, which is the leading cause of acute gastroenteritis in infants and young children in developing countries. By employing molecular cloning techniques and sequencing software, I investigated the mechanism by which the dsRNA editing enzyme ADAR1 gene expression is regulated during microbial infection in the presence of β-interferon (β-IFN). Given that ADAR1 catalyzes the deamination of adenosine to inosine and leads to RNA instability, I was highly interested in determining what fundamental elements of ADAR1 dictate its substrate selectivity. My work showed that β-IFN-stimulation controls ADAR1 promoter

usage and upregulates its alternative mRNA splice variants 7a and 7b, which have been shown to influence substrate specificity. I presented my findings at the **2012 U.S. Amgen Scholars Symposium**.

My experience with computational genomics at UCSF exposed me to the emerging need for the integration of biology with computer science and inspired me to pursue an internship in engineering. I was awarded the **2013 Howard Hughes Medical Institute (HHMI) Fellowship** to conduct independent research at the **California Institute of Technology**. Under the mentorship of **Dr. Morteza Gharib** I constructed a time-lapse imaging incubation system to examine mechanical strain on NIH/3T3 mouse embryonic fibroblasts (MEFs) *in vitro*. First, I designed an incubation chamber around a stereomicroscope to control for temperature, CO_2 content, humidity, and to provide an optimally buffered culture medium for the MEFs. Second, I configured a micromanipulator to induce regulated, measurable amounts of mechanical stress in order to prevent extreme strains on the MEFs. This research experience challenged me to integrate my knowledge of molecular signaling and bioengineering to creatively show that soft tissue may be regenerated independently of material scaffolds in the hopes of designing minimally invasive tissue regeneration procedures. My work resulted in a **first-author publication in the *Caltech Undergraduate Research Journal*[1]** and I was selected as a finalist for the **5th Annual Winter Enrichment Program (WEP) Undergraduate Poster Competition** at the **King Abdullah University of Science and Technology (KAUST)** in **Jeddah, Saudi Arabia**.

To develop a deeper understanding of cell signaling and membrane biology, I was awarded the **2014 Intramural Research Training Award** to pursue full-time research at the **National Institutes of Health (NIH)**. Under the mentorship of **Dr. Julie Donaldson** at the National Heart, Lung and Blood Institute (NHLBI), I investigated the physiological function of clathrin-independent endocytosis (CIE) as well as the mechanisms regulating the sorting of endocytosed proteins. I designed and implemented an RNAi screen of ~100 human de-ubiquitinating enzymes (DUBs) to identify mediators of CIE cargo sorting in HeLa cells using flow cytometry. Since DUBs remove ubiquitin and therefore rescue target proteins from degradation, we hypothesized that DUBs protect specific CIE-specific cargo from endolysosomal degradation as a specialized function to retain key nutrient transporter proteins – a function that may be exploited by cancer cells. I identified three DUBs that individually protect CD98, a nutrient transporter protein, from endolysosomal degradation. I presented my work at the **NHLBI Research Festival** and **Postbac Poster Day**, where I was awarded the **Outstanding Poster Award**. Now at UCB, I am collaborating with Dr. Donaldson to test the synergistic effect of our 3 hits on CD98.

Based on my profound interest in membrane biology, I am pursuing my PhD in the Department of **Molecular Cell Biology (MCB)** at the **University of California, Berkeley**

(UCB) and was accepted to **UCB's Chemical Biology Graduate Program (CBGP).** The interdisciplinary natures of MCB and CBGP offer a highly collaborative, dynamic research environment to work at the interface of biology and chemistry. In combination with the incredible resources of UCB, I trust that my extensive experience in membrane trafficking will enable me to successfully carry out my proposed NSF research project with **Dr. James Hurley**. Dr. Hurley is a world leader in membrane trafficking with expertise in Cryo-EM and X-ray crystallography, recently publishing the first crystal structure of the HIV/AIDS Nef-AP2 complex and pioneering a new direction for anti-HIV therapeutics. I am honored to partner with Dr. Hurley to further characterize the structural plasticity of endocytotic machinery.

Broader Impacts: I chose Grinnell because of its strong tradition of social justice. At Grinnell, I became an active member of the leadership team of **Grinnell Women in Science**. I advocated for the advancement of underrepresented groups in **STEM education** by hosting 4-hour scientific workshops on Saturday mornings for 12-16 year old girls. We taught short courses on biology and chemistry subjects, presented the work of inspiring female scientists, and led exciting science activities and demonstrations. This experience led me to question how I could reach children who could not access these outreach programs in developing countries and what I could do to help. Beginning in December 2011, I **co-founded** and **co-lead** the **Grinnell Chapter of Global Medical and Public Health Brigades**. We mobilized student volunteers to recruit physicians and dentists to host free clinics and nutritional literacy classes for ~350 community members in Las Delicias, **Honduras** in May 2012 and ~300 residents in Portuchada, Panama in March 2013. For our brigade to **Panama**, we received a **$100,000** grant from the Catholic Medical Mission Board for medications and vitamins. At each of these brigades, I led *Charla* workshops, which include short lectures and creative activities about the value of oral hygiene for children ages 2-12. I managed interactive, hands-on demonstrations that educated participants on important hygiene practices, such as brushing properly and flossing regularly. In August 2013, I **led** Grinnell's first public health brigade to build sanitation units for two families in El Hatillo, **Nicaragua**. To this day, our chapter has fundraised **$145,370** that has gone towards implementing sustainable measures to break the cycles of poverty and inspire the pursuit of education, such as the development of local schools and Community Health Worker training workshops to empower local leaders to sustain a consistent level of affordable health care and health literacy resources. I presented an ethical review of our work and contributions at the **2013 Heartland Global Health Consortium.**

My international volunteer experience led me to question how I could connect the benefits of basic research, including therapeutic agents, to those who need it most. This question inspired me to conduct a **Mentored Advanced Project at Grinnell with Dr. Eliza Willis in Global Health Policy**. I partnered with the Ministry of Health in Estelí, **Nicaragua** to show that non-profit organizations that empower local leaders can more sustainably improve the

medication delivery systems to rural, isolated communities. I saw the value of integrating global development with health policy to sponsor global initiatives for those who could benefit from my research the most. In addition to my global policy development experience, I engaged in public policy research opportunities to learn about additional ways to advocate for local disadvantaged children and motivate them to pursue science. In 2013, I worked closely with Team Nutrition and the Iowa State Department of Education to develop policies to combat child food insecurity. My research showed that the policy alternative to expand the nationwide initiative *Breakfast in the Classroom* would benefit the highest number of children in low-income communities in Iowa. I presented my findings at the **3rd Annual Peace and Conflict Studies Conference** and my work resulted in a **first-author publication** in the ***Journal of Peace and Conflict Studies*[2]. ** Moreover, I was elected to the **Student Educational Policy Committee** of Grinnell's Department of Biological Chemistry to advocate for strategic improvements in our curriculum to ensure STEM courses was accessible for college students from underrepresented backgrounds. I also conducted surveys to systematize student feedback for professor interim and tenure reviews in the hopes of enhancing the infrastructure for research and teaching within the department.

During my fellowship at the NIH, I was accepted to the **NIH Academy Fellows Program** where I discovered how to integrate **health disparities research** into my work as a future scientist. I launched a student-run organization that worked with Keys to Canaan, a 501(c)3 non-profit focused on the advancement of education for high-risk communities in South Washington, D.C. For our first project, we partnered with the D.C. Office of the Chief of Technology Officer (OCTO) to provide affordable internet and free programming classes to parents as part of the Digital Inclusion Initiative. My leadership team and I were able to recruit 15 families to take advantage of these opportunities. Through my volunteer experiences have learned that education empowers communities to inspire future generations to encourage positive change. Now at UC Berkeley, I aspire to continue my advocacy work, with a special focus on increasing access to STEM education.

Berkeley shares Grinnell's philosophy of social responsibility and action and offers an excellent setting to further my advocacy goals. Currently, I tutor math to high school students, ages 14-18, from low-income, high-risk areas of Oakland through an organization called **Upward Bound Math/Science Berkeley**, a partner of the national federal initiative: Pre-College TRiO Programs. I have had the opportunity to tutor students one-on-one and groups of 4-6. Also, I have become an active member of **Bay Area Scientists in Schools** (**BASIS**) to teach biology and chemistry in elementary schools. Bimonthly, we host scientific demonstrations for kindergarten students. Moreover, as an Amgen Scholar alumnus, I plan to tutor and mentor future summer UCB Amgen Scholars. Through my current and future volunteer initiatives, I hope to continue to empower disadvantaged communities, in the hopes of developing of a diverse, globally competitive STEM workforce.

Publications: [1] **Eshun-Wilson, L.**, H. Azizgolshani, M. Gharib. 2014. Tissue Expansion for Organ Construction and Tissue Regeneration: A Focus on Design and Instrumentation. *Caltech Undergraduate Research Journal*, 15, 9-16. [2] **Eshun-Wilson, L.** L. Van Pilsum-Bloom, D. Yong Shim, C. Lee, D. Hess, L. Lyons. 2014. Alleviating Child Food Insecurity in Iowa by Expanding the Breakfast in the Classroom Initiative. Journal of Peace and Conflict Studies, 1, 26-38.